Paper Summary

Blockchain-Based Adaptive Trust Management in Internet of Vehicles Using Smart Contract

Problem

Vehicles exchange safety and traffic messages, but an authenticated sender can still provide false information. A false congestion warning or an incorrect road-safety message can affect driving decisions. Singh and colleagues investigate how vehicles can maintain a consistent reputation history, reward useful reports and restrict repeated misbehavior without relying on one central trust database. [1]

Proposed approach

The paper places blockchain-supported trust management at the roadside infrastructure. Vehicles report suspicious behavior; roadside units analyze reports and update trust through smart contracts. The Traffic Authority registers participants, while a Certificate Authority supports identity credentials. Regional Authorities manage local infrastructure and short-term identities. The proposed architecture combines a global blockchain with regional shards so that local transactions need not all burden the global ledger. [1, Sections III–V]

Platform and workflow

The implementation uses Ethereum, Geth and Solidity, with JavaScript, web3.js and Node.js for interaction. The architecture describes Proof of Work for the global chain and an authoritative validator mechanism for regional shards. A reporting session requires at least four participants and a strict majority against the suspect. The narrative reduces the suspect's trust by one and temporarily blocks it for one minute; repeated misbehavior can cause hard revocation below zero trust. A successful reporter gains one trust point and five credit points, with two extra credits for the first reporter. [1, Algorithms 1–3]

Evaluation

The reported testbed contains four PCs: one Traffic Authority and three roadside units, plus a Raspberry Pi acting as a vehicle light node. It evaluates transaction handling, including throughput and execution time. The testbed establishes implementation feasibility; the broader sharded architecture should not be treated as a fully demonstrated large-scale deployment. The paper does not report a MultiChain or SUMO implementation. [1, Sections V–VI]

Conclusion and relevance

The conclusion explicitly leaves actual misbehavior detection and local plausibility checks for future work, including position, speed, heading and beacon-related checks. It also identifies privacy, AI-assisted detection and efficient consensus as future directions. The accompanying project addresses one narrow part of that gap: a Python position–speed consistency detector driven by SUMO, with evidence and decisions recorded in MultiChain. This is a classroom adaptation, not the authors' original implementation. [1, Section VII]

Reference: P. K. Singh, R. Singh, S. K. Nandi, K. Z. Ghafoor, D. B. Rawat and S. Nandi, IEEE Transactions on Intelligent Transportation Systems, accepted 2020. DOI 10.1109/TITS.2020.3004041.